

Too few, too many, or just right? Optimizing sample sizes for population-level inferences in animal tracking projects

Inês Silva^{1,*}, Christen H. Fleming^{2,3}, Michael J. Noonan^{4,5,6}, William F. Fagan⁷, Justin M. Calabrese^{1,7,8}

¹Center for Advanced Systems Understanding (CASUS), Helmholtz-Zentrum Dresden-Rossendorf (HZDR), Görlitz, Germany;

²Department of Biology, University of Central Florida, Orlando, Florida, USA;

³Smithsonian National Zoo and Conservation Biology Institute, Virginia, USA;

⁴Department of Biology, University of British Columbia Okanagan, Kelowna, Canada;

⁵Okanagan Institute for Biodiversity, Resilience, and Ecosystem Services, University of British Columbia Okanagan, Kelowna, Canada;

⁶Department of Computer Science, Math, Physics, and Statistics, University of British Columbia Okanagan, Kelowna, Canada;

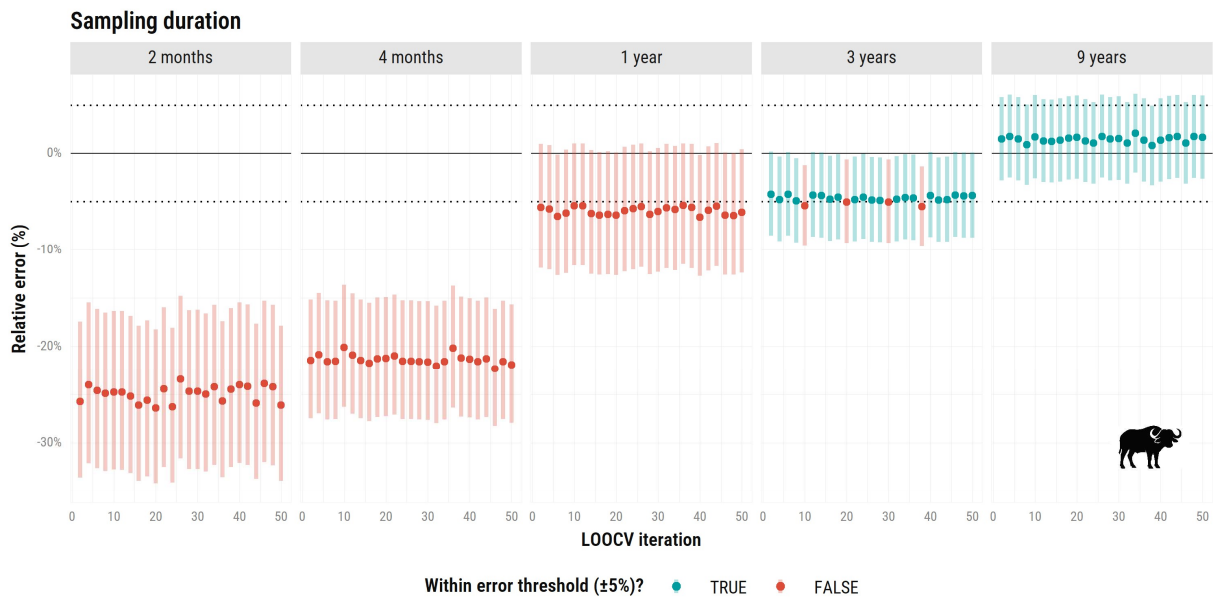
⁷Department of Biology, University of Maryland, Maryland, USA;

⁸Department of Ecological Modelling, Helmholtz Centre for Environmental Research – UFZ, Leipzig, Germany

***Corresponding author:** Inês Silva, **E-mail:** i.simoessilva@hzdr.de, imss.silva@gmail.com

S4. Leave-one out approach

18 Home range estimation



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20 **Figure S4.1.** Relative error (%) in home range area estimates across different sampling durations for simulated African buffalos (*Syncerus caffer*) for
21 a leave-one-out approach. Points represent the mean relative error (%) for one of 50 iterations (excluding a different individual each time), with its 95%
22 confidence intervals. Horizontal dotted lines indicate the $\pm 5\%$ error threshold. Blue points fall within this threshold; red points fall outside of it.



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24 **Figure S4.2.** Relative error (%) in home range area estimates across different sampling durations for simulated Mongolian gazelles (*Procapra*
25 *gutturosa*) for a leave-one-out approach. Points represent the mean relative error (%) for one of 50 iterations (excluding a different individual each
26 time), with its 95% confidence intervals. Horizontal dotted lines indicate the $\pm 5\%$ error threshold. Blue points fall within this threshold; red points fall
27 outside of it.

Speed & distance estimation



Figure S4.3. Relative error (%) in speed & distance estimates across different sampling intervals for simulated African buffalos (*Syncerus caffer*) for a leave-one-out approach. Points represent the mean relative error (%) for one of 50 iterations (excluding a different individual each time), with its 95% confidence intervals. Horizontal dotted lines indicate the $\pm 5\%$ error threshold. Blue points fall within this threshold; red points fall outside of it.



Figure S4.4. Relative error (%) in speed & distance estimates across different sampling intervals for simulated Mongolian gazelles (*Procapra gutturosa*) for a leave-one-out approach. Points represent the mean relative error (%) for one of 50 iterations (excluding a different individual each time), with its 95% confidence intervals. Horizontal dotted lines indicate the $\pm 5\%$ error threshold. Blue points fall within this threshold; red points fall outside of it.